Energy 232 (2021) 120995

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Multiobjective bilevel optimization algorithm based on preference selection to solve energy hub system planning problems

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ARTICLE INFO

Article history:
Received 8 August 2020
Received in revised form
23 April 2021
Accepted 17 May 2021
Available online 27 May 2021

Keywords:
Energy hub
Bilevel optimization
Multiobjective
Evolutionary algorithm
Preference

ABSTRACT

Energy hub system planning is a large-scale discrete multiobjective problem and it also belongs to a Stackelberg game. It is difficult to obtain a solution to this problem in a limited time through deterministic algorithms. In order to solve the above problems, a multiobjective bilevel optimization algorithm based on preference selection is proposed, which is divided into lower-level optimization and upper-level optimization. The preference selection mechanism can solve the uncertainty of the lower-level decision-making, and the trisection search method can improve the speed of the upper-level optimization. In the energy hub system planning problem, the upper-level optimizes the best capacity of energy equipment, and the lower-level optimizes the best combination of each energy carrier. Compared with other heuristic algorithms, the proposed method saves the computational time required to solve the problem. Compared with the commercial optimizer, the proposed method makes up for the defect that the commercial optimizer cannot solve the nonlinear discrete problem. The proposed method helps to solve the planning, design and operation scheduling problems of complex energy hub systems and multi-energy flow complementary systems. This method provides a theoretical basis for further research on the optimal scheduling of the entire life cycle of the energy hub system.

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1. Introduction

Traditional energy systems, such as electrical systems, natural gas systems, and cooling and heating system, are mostly designed, planned, and operated separately. Since lack of coordination among these conventional systems seriously compromises the entire energy system's economical-efficient performance. With the advancement of energy internet technology [1,2], the coupling relations of all kinds of energy (electricity, gas, heat, and cold) are significantly improved, and the interaction among energy supply, energy transport, and energy demand is continuously strengthened and highlighted [3]. Therefore, the coordination-operation of multiple energy resources has become an increasingly agenda and task in energy applications. In recent years, the integrated energy systems of cooling and heating equipment based on refrigerators, heat pumps, gas-fired boilers, and electric power and natural gas networks have gradually become a hot point in multi-energy systems research. Energy hub is an input-output dual-port model employed to character the energy supply, conversion, storage combined with coupling relationship between different types of energy and load demand in multi-energy systems [4]. The coupling matrix is introduced in the energy hub to reflect the relationship between the conversion, distribution, and storage of multiple energy sources. The energy hub model can coordinate and optimize

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https://doi.org/10.1016/j.energy.2021.120995
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different types of energy equipment and energy demand in multi-energy systems. However, the design of the energy hub system involves the coupling of different types of energy equipment. Simultaneously, capacity allocation and operation optimization control problems need to be considered to reduce the system's initial investment cost, operating cost, and carbon emissions. So above requirements make the optimization and management of the energy hub system a well-known and challenging problem.

At present, scholars have conducted different studies on the planning and operation of energy systems using the concept of energy hubs. The planning and operation of energy hub systems are mostly mixed-integer linear or mixed-integer nonlinear problems, and commercial optimizers such as CPLEX and GUROBI are mostly used to solve them. In this research, Schulze et al. [5] proposed an energy hub modeling framework to determine the optimal power flow problem for integrated energy systems and multi-energy carriers. Taking into account the impact of energy prices and carbon dioxide tax rates on the performance of the energy hub system, Ren et al. [6] proposed a mixed-integer linear model of the energy hub to analyze the planning and operation modeling of the integrated energy system, aiming to minimize the total energy demand. Ahmadisedigh et al. [7] established a mixed-integer linear model of the energy hub system from the perspective of cost and greenhouse gas emissions, and used CPLEX to optimize the system design and operation problems under different heating and cooling load combinations. Guler et al. [8] proposed regional energy hubs, established a coupling model for regional energy hubs, and used CPLEX to optimize the operating costs and greenhouse gas emissions of the regional energy hub system. Brahman et al. [9] used the CPLEX optimizer to optimize the energy system multiobjective optimization problem with the best equipment capacity, maximum return on investment, and minimum carbon emissions as the objectives. Maroufmashat et al. [10] used the method of weighted summation of objective functions to convert a dual-objective energy hub planning problem to minimize cost and greenhouse gas emissions into a single-objective problem and finally solved it with the CPLEX optimizer. In addition to economic and environmental considerations, with the further utilization of renewable energy, the integration of renewable energy and the grid is also a consideration for the planning and operation of the energy hub system. Therefore, Rakipour et al. [11] proposed a comprehensive energy system (CES) with the concept of an energy hub, including electricity, heating and cooling, renewable energy (RERS), and a demand response plan for heat and power. The system aims at maximizing the profit of the energy hub, uses GAMS to model the energy system to obtain a mixed-integer linear model, and finally uses the Cplex optimizer to solve it. Jamalzadeh et al. [12] proposed a mixed interval stochastic framework and established a robust mixed-integer linear programming of the energy hub, and then the CPLEX optimizer solved the programming problem. In their research, the energy system optimization problems that were proposed largely rely on optimizers such as CPLEX. But these optimizers can only solve some problems (e.g, linear problems, continuous derivable problems, etc.), and will be powerless in the face of complex nonlinear discrete problems.

In addition, with the continuous development and promotion of heuristic algorithms, more and more scholars use various single-stage heuristic algorithms to study the planning and operation of energy hub systems [13,14]. Huang et al. [15] aimed at the photovoltaic-heat pump-heat storage-electric vehicle coupled energy hub system, aiming at the maximum self-consumption in the economic life, using genetic algorithms to optimize the capacity and location of photovoltaic modules. Under the influence of heat storage, the penetration rate of electric vehicles and the possibility of power sharing, how the optimal photovoltaic capacity, self-sufficiency, and self-consumption power affect the size, location and performance indicators of the photovoltaic system are studied. Chen et al. [16] established a combined cooling, heating and power multi-energy system using wind and solar energy. Based on considering the investment cost, primary energy consumption, and carbon dioxide emissions, the particle swarm optimization (PSO) is used to optimize the multi-energy distribution problem in the multi-energy system. Eladl et al. [17] proposed a multi-energy hub optimized operation and configuration method. Considering the uncertainty of wind energy and photovoltaics, a multiobjective optimization problem of maximizing social welfare and minimizing carbon dioxide emissions is established, and genetic algorithm (GA) is used to solve the optimization problem. Huo et al. [18] considered the operation optimization of the energy hub system with batteries, and proposed a PSO combined with the internal node method to solve the energy storage system scheduling problem, but this method can only approach the global optimal value. Although the single-stage heuristic algorithm can get rid of the optimizer to solve the energy hub system planning problem, it cannot be solved in the face of particularly complex energy system planning problems. As energy system model designs are becoming more complex, the problem that usually needs to be optimized is a large-scale problem. There is a poor calculation efficiency and poor optimization results when using a single-stage heuristic algorithm to solve large-scale problems.

In summary, there are currently two mainstream methods for energy system optimization problems: solving with commercial optimizers (e.g, CPLEX and GUROBI) and solving with single-level heuristic algorithms (e.g, GA and PSO). As energy system models become more complex, the two mainstream methods will no longer be the best choice. In certain problems, the above methods may not be able to solve them. If we use meta-mathematics method to solve, we may face the problem of indirect objective function. If a deterministic computer algorithm is used to solve the problem, the time complexity of the algorithm is very high. If you use a commercial solver to solve the problem, it is difficult to obtain the global optimal solution and the calculation time is long. If a single-stage heuristic algorithm is used to solve this problem, due to the high dimensionality of the decision variables, it faces failure to converge. To solve the above problems, a multiobjective bilevel optimization algorithm based on preference selection is proposed to solve the energy hub system planning problem. Energy hub system planning is a multiobjective mixed integer planning problem, which also belongs to the Stackelberg game [19,20]. Design a bilevel optimization algorithm for the problem. The upper-level uses the trisection search method to optimize the optimal capacity of the energy equipment, and the device capacity value is passed to the lower-level optimizer as a parameter. The lower-level uses evolutionary algorithms to optimize the best operating power combination of each energy device to obtain multiple solutions, and then select one of the solutions based on preference. The main contributions of this work are summarized as follows:

(1) Since the dimension of decision variables in the energy hub system planning problem is as high as tens of thousands of dimensions, it is not realistic to use existing evolutionary algorithms to simultaneously optimize all decision variables. For large-scale problems (problems with the high dimensionality of decision variables), decision space dimensionality reduction is a common method to solve large-scale problems. The proposed bilevel optimization algorithm separates the decision variables and puts them into the upper and lower-level optimizers for optimization (that is, the upper-level optimizes the capacity of the device, and the lower-level optimizes the real-time operating power of the

device), so as to achieve the purpose of reducing the dimensionality of the decision space.

(2) When the lower-level optimizer optimizes the equipment's real-time operating power throughout the year, if it is optimized once per hour, it needs to be optimized 8760 times throughout the year, which requires a lot of time and cost. We propose an optimization strategy based on typical moments, and the results at atypical moments are obtained by Lagrange interpolation function fitting. After many optimizations, the lower-level will pass the whole year's optimization results to the upper-level.

(3) To solve non-deterministic decision-making in the lower and middle layers of multiobjective bilevel optimization, we propose a decision-making mechanism based on preference choice. The optimization result of the lower-level optimizer is a set of non-dominated solutions, which is selected as the final optimization result of the lower-level optimizer according to the preference information and passed to the upper-level optimizer.

(4) In the case of this paper, the upper-level optimizer we designed uses a trisection search to search for the best capacity value of the best heat pump. The time complexity of the search algorithm is $O(log_3 n)$, so it can significantly improve the entire bilevel optimization algorithm's computational efficiency.

The rest of the paper is organized as follows. In Section 2, we briefly introduced the related concepts and methods used to propose the algorithm. In Section 3, the algorithm framework and program implementation are described in detail. In Section 4, the proposed algorithm is applied to an energy hub system planning case, and the simulation results are analyzed. In Section 5 summarizes the paper.

2. Related concepts and methods

2.1. Multiobjective bilevel optimization

Bilevel optimization is widely used in practical problems, such as problems setting up the charges (formulate transportation policies [21], determine highway tolls [22], and formulate optimal tolls [23]), environmental economics (packet analysis [24], agricultural environment policy [25]), machine learning (feature selection [26] and classification model for complex control tasks [27]), and others (optimization of energy supply systems [28]). Most bilevel optimizations can be represented by the upper-level and lower-level optimization frameworks, and each decision variable of the upper-level requires the response of the lower-level decision. The upper and lower-level of a bilevel optimization model have their objective functions and constraints, and the lower-level is nested in the upper-level [29]. The lower-level parameterizes the optimization task and is a constraint for superior optimization. Multiobjective bilevel optimization is a nested optimization problem involving two levels of multiobjective optimization tasks [30]. The structure of a multiobjective bilevel problem demands that only the Pareto-optimal solutions to the lower-level optimization problem be considered as feasible solutions for the upper-level optimization problem. There are two classes of variables in a bilevel optimization problem; namely, the upper-level variables $x_u \in X_U \subset \mathbb{R}^n$, and the lower-level variables $x_l \in X_L \subset \mathbb{R}^m$. The lower-level multiobjective problem is solved with respect to the lower-level variables, x_l, and the upper-level variables, x_u, act as parameters to the optimization problem. Each x_u corresponds to a different lower-level optimization problem, leading to a different Pareto-optimal front. The upper-level problem is optimized with respect to both classes of variables, $x = (x_u, x_l)$.

For the upper-level objective function $F : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^p$ and lower-level objective function $f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^q$, the bilevel problem is given by

$$\begin{aligned} &\min_{x_u \in X_U, x_l \in X_L} F(x_u, x_l) = \left(F_1(x_u, x_l), \ldots, F_p(x_u, x_l)\right) \\ &\quad \textit{subject to} \\ &\qquad x_l \in \underset{x_l}{\operatorname{argmin}} \{ f(x_u, x_l) = \Big(f_1(x_u, x_l), \ldots, f_q(x_u, x_l)\Big) : \\ &\qquad\qquad g_j(x_u, x_l) \le 0, j = 1, \ldots, J \Big\} \\ &\qquad G_k(x_u, x_l) \le 0, k = 1, \ldots, K; \end{aligned} \tag{1}$$

where $G_k : X_U \times X_L \to \mathbb{R}, k = 1, \ldots, K$ denotes the upper-level constraints, and $g_j : X_U \times X_L \to \mathbb{R}$ represents the lower-level constraints, respectively. Equality constraints may also exist that have been avoided here for brevity.

2.2. Multiobjective evolutionary algorithm

Multiobjective evolutionary algorithms are widely used in multiobjective non-polynomial time complexity (NPC [31]) problems. A multiobjective evolutionary algorithm is good at solving nonlinear and discrete problems. Evolutionary algorithms perform well in many areas, such as problems that are large scale, constraints, preferences, many-objective, multi-modal, or dynamics [32]. Multiobjective evolutionary algorithms can be divided into the following categories: multiobjective evolutionary algorithms based on Pareto advantage (NSGA-II [33], SPEA-II [34]), multiobjective evolutionary algorithms based on reference points (NSGA-III) [35], multiobjective evolutionary algorithms based on decomposition (MOEA\D) [36], and multiobjective evolutionary algorithms based on indicators (HypE) [37]. Evolutionary algorithms based on reference points, reference vectors, and indicators are all proposed to solve many-objective problems, and their design ideas and execution processes are relatively complicated. However, the paper only needs to solve a bi-objective problem, so the lower-level optimizer uses an evolutionary algorithm based on the Pareto advantage. The lower-level makes deterministic decisions through the preference selection mechanism and only transmits a solution to the upper-level optimizer, so the lower-level optimizer should use a more convergent algorithm. Finally, we improved NSGA-II for the problem and embedded it in the lower-level optimizer. The following is a brief introduction to the concepts related to evolutionary algorithms used in this paper, such as multiobjective optimization problems, Pareto optimal solutions, Pareto dominance, Pareto optimal sets, and a Pareto optimal frontier.

Multiobjective optimization problems (MOPs) can be mathematically defined as:

$$\begin{cases} \textit{min } F(X) = (f_1(X), f_2(X), \ldots, f_m(X)), \\ \textit{subject to } X \in \Omega, \end{cases} \tag{2}$$

where $X = (x_1, x_2, \ldots, x_n)$ is an n-dimensional decision variable vector from the decision space Ω; $F(X)$ is an objective function vector that consists of m conflicting objective functions [38].

Unlike single-objective optimization, a set of trade-off solutions, termed Pareto optimal solutions, is expected to be found for MOPs. Let $X_1, X_2 \in \Omega$; X_1 is said to dominate X_2, denoted by $X_1 \prec X_2$, if and only if $f_i(X_1) \le f_i(X_2)$ for each $i \in \{1, \ldots, m\}$ and $f_j(X_1) < f_j(X_2)$ for at least one indicator $j \in \{1, \ldots, m\}$; if all X from Ω cannot dominate X_1, we call X_1 a nondominated or Pareto optimal solution. The set of all Pareto optimal solutions is called the Pareto optimal set (PS), and

the set of all the Pareto optimal objective vectors is called the Pareto optimal front (PF). Any improvement in one objective of a Pareto optimal solution is bound to deteriorate at least one other objective [38].

3. Propose algorithm framework

3.1. Lower-level optimization

The lower-level optimizer's main job is to optimize the real-time operating power of each device throughout the year. It is not realistic to optimize the real-time operating power of the equipment throughout the year at one time. In order to solve the above problems, a method of optimizing the operating power of the equipment per hour for a total of 8760 times was proposed. Although the above method solves the problem of not being able to optimize, it has too high a time cost and too low computational efficiency. We further improved the lower-level optimizer, keeping the idea of multiple optimizations but not optimizing 8760 times. The final method is only to optimize the results at typical moments, and a fitting function approximates the results at atypical moments. In the multiobjective bilevel optimization, due to non-deterministic decision-making at the lower-level, we propose a method based on preference selection. The lower-level optimizer obtains the annual optimization result after multiple optimizations, that is, the operation combination plan of each equipment in the system under a certain equipment capacity. The lower-level optimization obtains multiple sets of solutions, so one solution needs to be selected through preference and passed to the upper-level optimizer.

Algorithm 1
NSGA-II-CDP

Input: Equipment capacity value, $Upper_{data}$; Typical moment, T;
Output: Operation combination scheme of each device at time t, $Tlower_{data}$;
1: **function** NSGA-II-CDP($Upper_{data}$, T)
2: Set parameters $Upper_{data}$ and T;
3: $P \leftarrow InitPop()$;
4: *CalculateFitnessAndConstraintDegree*(P);
5: $Q \leftarrow MakeNewPop(P)$; /*Selection, crossover, mutation operations*/
6: $R \leftarrow MergePop(P, Q)$;
7: $F \leftarrow NondominatedSort(R)$; /*Use ENS [39]*/
8: *CalculateCrowdingDistance*(F);
9: *EstablishPartialOrderRelationships*(R);
10: $P \leftarrow SelectPop(R)$; /*Choose n in $2n$*/
11: **if** *termination condition not fulfilled* **then**
12: *goto step* 2; /*Line 2*/
13: **else**
14: **return** $Tlower_{data}$
15: **end if**
16: **end function**

3.1.1. Improve the NSGA-II

NSGA-II [33] is a classic multiobjective evolutionary algorithm, which is efficient and straightforward and has an excellent effect on solving bi-objective problems. Since the original NSGA-II cannot handle constraints, we have added a constraint handling mechanism. The algorithm's crossover and mutation operators also made targeted improvements. Algorithm 1 is an improved program of NSGA-II. The first step is to set the algorithm parameters and randomly generate an initial population P (lines 2–3). The second step is to calculate individuals' fitness in the population and the degree of constraint violation (line 4). The third step is for the parent population to perform selection, crossover, and mutation operations to generate the offspring population Q (line 5). The fourth step is to merge the parent and offspring populations to obtain the population R and perform a fast non-dominated sorting on the population R, and then calculate the crowding distance of the individuals in the population R, finally, the partial ordering relationship of the population R is established through the non-dominated ranking level and the crowding distance (lines 6–9). The fifth step selects n individuals from the population R to form the population P according to the partial order relationship. The sixth step is to determine whether the algorithm meets the termination condition. If it is satisfied, it jumps to line 2. If it is satisfied, the function returns the optimization result ($Tlower_{data}$).

3.1.2. Lagrange Interpolation

Lagrange interpolation can fit a straight line to a given known point, and then use the straight line to approximate other positions. The mathematical formula can be expressed as:

Given a set of $k+1$ data points $(x_0, y_0), \ldots, \left(x_j, y_j\right), \ldots, (x_k, y_k)$, where no two x_j are the same, the interpolation polynomial in the Lagrange form is a linear combination $L(x) = \sum_{j=0}^{k} y_j \ell_j(x)$ of Lagrange basis polynomials:

$$\ell_j(x) = \prod_{0 \le m \ne j} \frac{x - x_m}{x_j - x_m} = \frac{(x-x_0)}{(x_j-x_0)} \cdots \frac{(x-x_{j-1})}{(x_j-x_{j-1})} \frac{(x-x_{j+1})}{(x_j-x_{j+1})} \cdots \frac{(x-x_k)}{(x_j-x_k)}, \tag{3}$$

where $0 \le j \le k$. Note how, given the initial assumption that no two x_j are the same, then(when $m \ne j$) $x_j - x_m \ne 0$, so this expression is always well-defined. The reason pairs $x_i = x_j$ with $y_i \ne y_j$ are not allowed is that no interpolation function L such that $y_i = L(x_i)$ can exist; a function can only get one value for each argument x_i. On the other hand, if also $y_i = y_j$, then those two points would actually be one single point. The Lagrange interpolation algorithm used in this paper is Algorithm 2.

Algorithm 2
Lagrange Interpolation

Input: Interpolating point, ($sample_x$, $sample_y$);
Output: Lagrange interpolation function, L;
1: **function** LAGRANGE($sample_x$, $sample_y$)
2: $NumPoint \leftarrow len(sample_x)$
3: $l \leftarrow [1]*num_point$
4: $L \leftarrow 0$
5: **for** $i = 0$; $i < NumPoint$; $i++$ **do**
6: **for** $j = 0$; $j < NumPoint$; $j++$ **do**
7: **if** $i == j$ **then**
8: *continue*
9: **end if**
10: $l[i] \leftarrow l[i]*(x - sample_x[j])/(sample_x[i] - sample_x[j])$
11: **end for**
12: $L \leftarrow L + l[i]*sample_y[i]$
13: **end for**
14: **return** L
15: **end function**

3.1.3. Deterministic decision based on preference

In the bilevel optimization model, upper-level optimization must obtain the deterministic solution of lower-level optimization. Therefore, the deterministic decision of the lower-level is very important, but the deterministic decision of the lower-level is usually a difficult point. The research on lower-level deterministic decisions has attracted the attention of scholars (ref [30]). Lower-level decisions can be divided into *DeterministicDecisions* [40] and *DecisionUncertainty* [41]. The following briefly introduces the difference between these two kinds of decision-making; deterministic decision means that the lower-level optimizer can decide the lower-level optimal solution, usually a single-objective problem or a problem that can directly obtain an optimal solution. Uncertainty decision means that the lower-level optimizer cannot decide the lower-level optimal solution. For example, in a multiobjective problem, it is a non-dominated solution set that makes it impossible to make a decision. Obviously, we all hope that lower-level decisions are deterministic, but we often encounter situations where lower-level decisions are uncertain. In order to solve uncertain decisions, we propose deterministic decisions based on preference. In the multiobjective problem, the lower-level optimization obtains a non-dominated set, which will form a PF. By using preference information to choose a solution, the purpose of deterministic decisions at the lower-level is achieved.

In order to further describe the deterministic decision-making based on preferences, Figs. 1 and 2 are given. In Fig. 1, the lower-level optimization has obtained five lower-level PFs (that is, PF1, PF2, PF3, PF4, PF5). Each lower-level PF is mapped to an area on the upper-level PF. It can be found that the solution on the same PF in the lower-level is not must correspond to the same PF on the upper-level. What we can only guarantee is that the five upper solutions of A, B, C, D, and E must be in the corresponding mapping area. Therefore, the deterministic decision at the lower-level is a prerequisite to ensure the execution of the upper-level optimization. In Fig. 2, the preference domain (i.e., a, b, c, d, e) is introduced, and only one solution is selected from the preference domain when making decisions at the lower-level, so the mapping to the upper PF is a solution instead of a region.

3.2. Upper-level optimization

In the energy hub system planning problem, the upper-level is mainly to optimize the heat pump's optimal capacity. In this paper's case, the upper-level only needs to optimize one variable, so thought of using an efficient search algorithm. It is widely known that binary search is an efficient computer search algorithm, but it is only suitable for searching in ordered arrays, and mapping to function search can only solve the solution of monotonic functions. This paper proposed a trisection search method (TSS) with time complexity $O(log_3 n)$ for convex functions. To better explain the principle of the algorithm, Fig. 3 shows the process of finding the minimum value of the convex function through the TSS. In Fig. 3(a), the ordinate Y is the objective function's value, and the abscissa X is the variable to be searched. The TSS algorithm is described in detail as follows:

(1) The search interval of the initialization variable x is [xl, xr];

(2) Find the trisection points midl, midr of the interval [xl, xr], which can be expressed as the x value corresponding to the red point in the Fig. 3(a);

(3) Calculate the objective function values at the points midl and midr. If midl $\prec$ midr, update the search interval [xl, xr] to [xl, $midr$], if midr $\prec$ midl, update the search interval [xl, xr] to [$midl$, xr];

(4) The search interval of the variable x is updated to [xl, xr] as shown in the Fig. 3(b). Operation step (2) obtains the trisection points (i.e., the x value corresponding to the blue point in the Fig. 3(b)), operation step (3) updates the search interval [xl, xr] again. The yellow point in the Figure 3(b) is the next trisection point;

(5) Repeat steps (2–4) until the interval length is less than the preset size, and get the minimum value of the convex function;

The TSS procedure is described in the Algorithm 3. The program

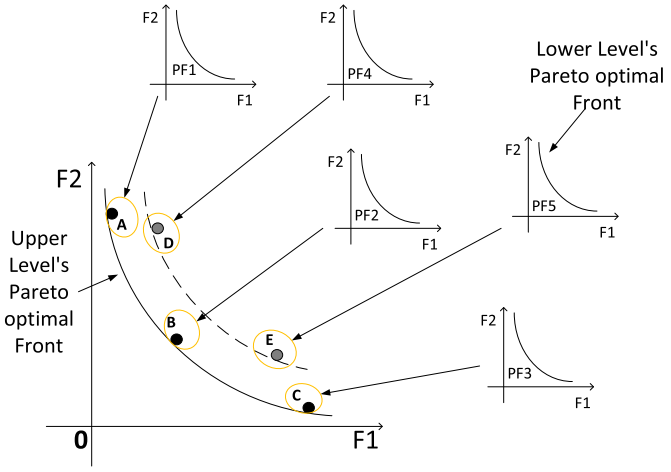


Fig. 1. Schematic diagram of uncertain decision making.

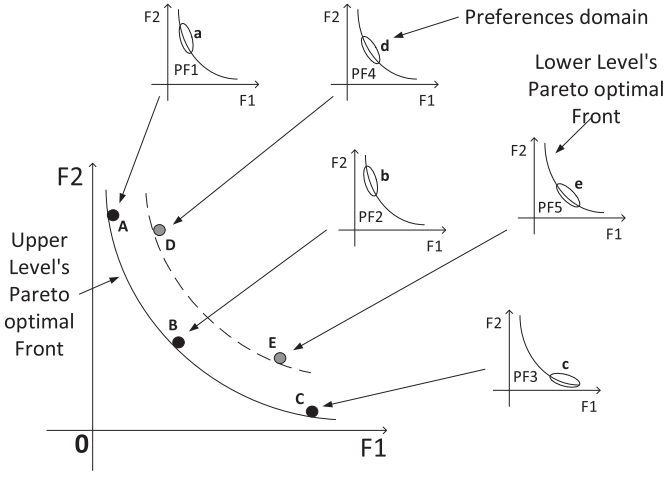


Fig. 2. Schematic diagram of deterministic decision based on preference.

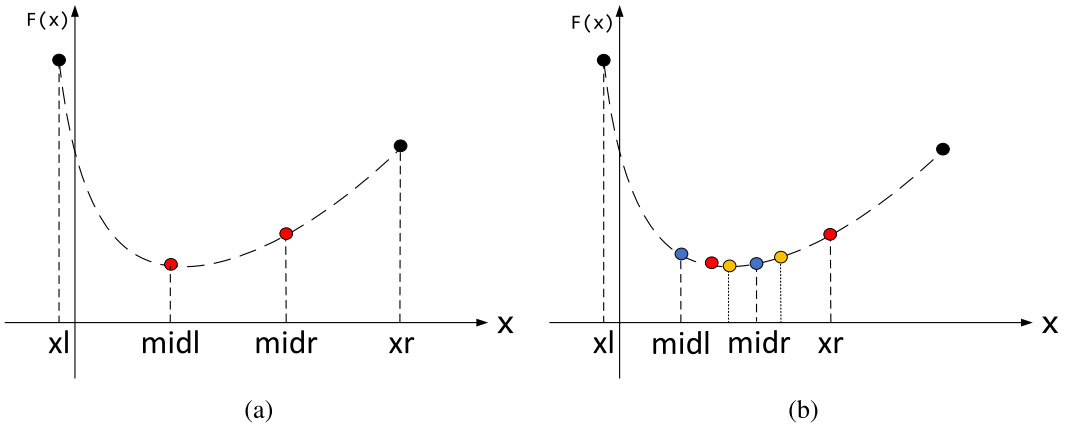


Fig. 3. The principle diagram of the TSS.

inputs the convex function $F(x)$, the search interval $[xl, xr]$ and the last interval length value *len*. The program outputs the optimal solution and its objective function value. Lines 3–4 are to calculate the trisection points *midl* and *midr*. The *LowerOptimization* function is a lower-level optimization function used to optimize the solution in a typical moment. The *Lagrangeroximate* function is the lower-level fitting function used to fit the annual optimization solution. Lines 5–10 mainly obtain the lower-level optimized solution and form the solution vector X_u_midl and X_u_midr together with the upper-level solution. $\prec$ is the symbol of Pareto's dominance relationship, which is defined in Section 2. Lines 11–18 compare the Pareto dominance of the solution vector and update the search interval. Lines 20–26 are to obtain the solution vector X_u_mid and three different preference solutions *resultL*, *resultM* and *resultR*. Lines 27–28 select a solution based on preference information, and the function returns this solution.

3.3. Integrated framework

This paper proposes a preference-based multiobjective bilevel optimization framework. The lower-level optimization is mainly composed of multiobjective evolutionary algorithms and fitting functions. The upper-level optimization uses TSS and uses the Pareto dominance relationship to judge the lower-level optimized solution. For related concepts and sub-algorithms used in the algorithm framework, please refer to Section 2 and Section 3. In Algorithm Framework 4, the input parameters are the user's preference information *prefer* and the number of typical moments *n*. The lower-level optimizer *LowerOptimization* optimizes by calling the Algorithm 1, and obtains the optimal solution $solution_{lower}$ (lines 1–6) at *n* typical moments. It is worth noting that we can use multi-threading technology to convert lower-level optimization into parallel optimization, which can greatly shorten the optimization time. Due to the very large scale of the problem, we only optimized the solutions at *n* typical moments, and then approximated the solutions at other atypical moments through the *Approximate* function (calling the Algorithm 2), and finally got the optimized solutions at all moments $solution_{approximate}$(lines 7–12). After the upper-level optimizer *UpperOptimization* sets the preference information and embeds the *LowerOptimization* and *Approximate* functions, it will call TSS (Algorithm 3) to optimize directly, and get the final solution *result* (lines 13–16). Finally, the solution is output, and the *view* function visualizes the solution (lines 17–19).

4. Algorithm application

The energy hub system studied in this paper can be described as shown in Fig. 4. The cooling water generated by chillers (*ch*) circulates in the building, so it serves to cool the building. The heat extracted from the building is transferred to the cold loop and is eventually dissipated to the environment in the cooling tower connected to the cooler, thereby becoming waste heat. The heat is provided to the building through steam or hot water network, which constitutes the second loop. Consider two types of boilers based on energy (i.e., electricity boilers (*eb*) and natural gas boilers (*fb*), respectively). Once the heat is absorbed from the steam or hot water, the condensate will return to the boiler. Generally, these two circuits (for example, chilled water and steam or hot water) are considered to be independent. The heat pump (*hp*) recovers heat from the cooling cycle, which can simultaneously supply heat to the building and produce chilled water.

Since there are no local generators or power plants in the system under consideration, the grid should directly meet the power demand. In the current model (Fig. 4), the power demand is divided into four parts: chillers, heat pump, electric boiler and the additional power demand (L_e). The cooling load L_c is satisfied by the chillers and heat pump, and the heat load L_h is satisfied by the heat pump and two types of boilers. The different loads identified in the Fig. 4 (i.e., L_e, L_h and L_c) are satisfied by two available energy sources, namely the grid (P_e) and natural gas (P_f). Since L_e, L_h and L_c change over time, so do P_e and P_f.

Algorithm 3
Trisection Search

Input: Convex function, $F(x)$; Value interval of x, $[xl, xr]$; Accuracy, len;
Output: Optimal solution and its objective function value, $(x_{min}, F(x_{min}))$;
function TSS($F(x)$, xl, xr)
while $xr - xl > len$ **do**
$midl \leftarrow xl + (xr - xl)/3$;
$midr \leftarrow (midl + xr)/2$;
$X'_midl \leftarrow LowerOptimization(midl)$;
$X_l_midl \leftarrow LagrangeApproximate(X'_midl)$;
$X'_midr \leftarrow LowerOptimization(midr)$;
$X_l_midr \leftarrow LagrangeApproximate(X'_midr)$;
$X_u_midl \leftarrow [midl, X_l_midl]$;
$X_u_midr \leftarrow [midr, X_l_midr]$;
if $X_u_midl \prec X_u_midr$ **then**
$xr \leftarrow midr$;
end if
if $X_u_midr \prec X_u_midl$ **then**
$xl \leftarrow midl$;
else
break;
end if
end while
$mid \leftarrow (midl + midr)/2$;
$X'_mid \leftarrow LowerOptimization(mid)$;
$X_l_mid \leftarrow LagrangeApproximate(X'_mid)$;
$X_u_mid \leftarrow [mid, X_u_mid]$;
$resultL \leftarrow (X_u_midl, F(X_u_midl))$
$resultM \leftarrow (X_u_mid, F(X_u_mid))$
$resultR \leftarrow (X_u_midr, F(X_u_midr))$
Select *result* by preference in {*resultL*, *resultM*, *resultR*};
return *result*
end function

Algorithm 4
Algorithm Framework

Input: Preference information, *prefer*; Typical moment number, n;
Output: The optimal solution to the problem, $solution_{data}$;
function LOWEROPTIMIZATION($variable_{upper}$)
for $i = 0$; $i < n$; $i++$ **do**
Algorithm 1/*NSGA-II-CDP*/
end for
return $solution_{lower}$/*All the solutions in a typical moment*/
end function
function APPROXIMATE($solution_{lower}$)
for $i = 0$; $i < n - 1$; $i++$ **do**
Algorithm 2/*Lagrange Interpolation*/
end for
return $solution_{approximate}$/*All solutions at all moments*/
end function
function UPPEROPTIMIZATION(*prefer*, *LowerOptimization*, *Approximate*)
Algorithm 3/*TSS*/
return *result*
end function
result ← *UpperOptimization*(*prefer*, *LowerOptimization*, *Approximate*);
output(*result*);
view(*result*); /*Output result simulation diagram*/

4.1. Upper-level model

4.1.1. Objective function

The upper-level model of the energy hub problem has two optimization objectives, namely, that the total cost of the energy hub system throughout the year be low, and that the total carbon dioxide emissions of the energy hub system throughout the year be low. TSS is used to optimize the upper model. The mathematical expression of the objective function is

$$F_1 : \min_{Q_{HP,\max},C_{op}} C_{tot} = \phi_{HP} Q_{HP,\max} + \frac{(1+i)^{n-1}}{i(1+i)^n} C_{op}, \tag{4}$$

$$F_2 : \min_{CO_{2,e},CO_{2,f}} CO_{2,\text{tot}} = CO_{2,e} + CO_{2,f}, \tag{5}$$

where C_{tot} is the total cost of the energy hub system throughout one year; $\phi_{HP} Q_{HP,\max}$ is the initial cost of the heat pump; $n = 10$ is the operating period of the energy hub; i is the growth rate of the operation and maintenance of the energy hub; C_{op} is the annual operating cost provided by the lower-level; $\phi_{HP} = 1218\ yuan/kWh$ is the price per unit heat pump capacity; $Q_{HP,\max}$ is the capacity of the heat pump; $Q_{HP,\max}$ is the upper-level decision variable; $CO_{2,\text{tot}}$ is the total carbon dioxide emissions of the energy hub system throughout a year; $CO_{2,e}$ is the carbon dioxide emissions of the electric power equipment; $CO_{2,f}$ is the carbon dioxide emissions of the natural gas boiler, and both are provided by the lower-level.

4.1.2. Constraint

After the upper model obtains the power of each device from the lower model, the capacity of the heat pump is optimized. The constraints of upper-level optimization are as follows:

$$st. \begin{cases} Q_{HP,\max} \geq 0, \\ Q_{HP,\max} \leq Qmax_{HP,\max}, \end{cases} \tag{6}$$

where $Qmax_{HP,\max} = 5454.55$ kW is the maximum capacity of the theoretical heat pump, that is, the upper limit that can be taken when optimizing the heat pump capacity, provided by the lower model.

4.2. Lower-level model

4.2.1. Objective function

The lower model uses the multiobjective evolutionary algorithm NSGA-II-CDP for optimization, which includes two optimization objectives. The first objective is to have a low total operating cost of the energy hub system throughout one year, and the second objective is to have low total emissions from the energy hub system throughout one year. The mathematical formula can be described as:

$$F_1 : \min_{P_e(t),P_f(t)} C_{op} = C_{op,e} + C_{op,f}; \tag{7}$$

$$C_{op,e} = \begin{cases} \phi_{e,\text{ night}} \sum_{t=1}^{N} P_e(t)\Delta t, & t \in [19,24); \phi_{e,day} \sum_{t=1}^{N} P_e(t)\Delta t, \quad t \in [0,19); \end{cases} \tag{8}$$

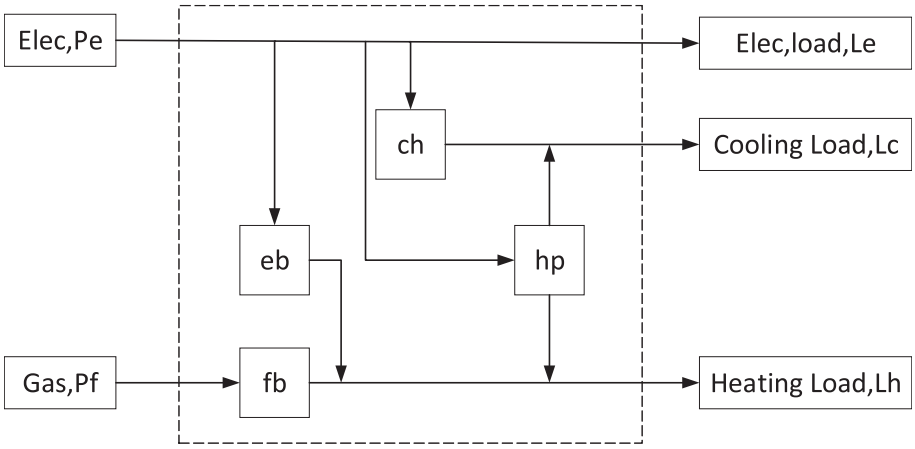


Fig. 4. Hub representation of a heating and cooling network including electrical and natural gas boilers (*eb* and *fb*), chillers (*ch*) and heat pumps (*hp*).

$$C_{op,f} = \phi_f \sum_{t=1}^{N} P_f(t)\Delta t; \tag{9}$$

$$F_2 : \min_{P_e(t),P_f(t)} CO_{2,tot} = CO_{2,e} + CO_{2,f}; \tag{10}$$

$$CO_{2,e} = \lambda_e \sum_{t=1}^{N} P_e(t)\Delta t; \tag{11}$$

$$CO_{2,f} = \lambda_f \sum_{t=1}^{N} P_f(t)\Delta t; \tag{12}$$

$$P_e = P_{e,EH} + P_{e,HP} + P_{e,CH} + L_e. \tag{13}$$

In these equations, C_{op} is the total operating cost of the energy hub system throughout the year; $C_{op,f}$ is the total annual operating cost of natural gas; $C_{op,e}$ is the total annual operating cost of electricity; $\phi_f = 0.316$ (*yuan*/*kWh*) is the price of natural gas per unit, and $\phi_{e,night} = 0.4$ (*yuan*/*kWh*) is the price per unit of electricity during peak non-use period [22:00–7:00); $\phi_{e,day} = 0.6$ (*yuan*/*kWh*) is the price per unit of high electricity consumption period [7:00–22:00); $P_f(t)$ is the total power of natural gas equipment at the t moment; $P_e(t)$ is the total power of electric equipment at time t, $\triangle t = 1\ h$ is the unit time; $N = 8760$ is the total time of the year; P_e is the total electric power; $P_{e,EH}$ is electric boiler power; $P_{e,HP}$ is heat pump power; $P_{e,CH}$ is refrigerator power, and L_e is user extra electric power. The above constant parameters are by the Hunan power grid standard in China. $CO_{2,tot}$ is the total emissions from the energy hub system throughout the year; $CO_{2,e}$ is the total annual waste emissions of electricity; $CO_{2,f}$ is the total annual natural gas emissions; $\lambda_e = 0.02072$ (*kg*/*kWh*) is the carbon dioxide emissions per unit of electric power; $\lambda_f = 0.17644$ (*kg*/*kWh*) is per unit carbon dioxide emissions from natural gas.

4.2.2. Constraint

The decision variables of the lower optimization model include electric boiler power ($P_{e,EH}$), natural gas boiler power (P_f), chillers power ($P_{e,CH}$), and heat pump power ($P_{e,HP}$). These variables must meet some constraints in practical problems. They are expressed as the following equation:

Cold and heat load function

$$\begin{cases} L_h = A_h + A_h \sin\left(2\pi \dfrac{t}{\tau_y} + \dfrac{\pi}{2}\right), \\ L_c = A_c + A_c \sin\left(2\pi \dfrac{t}{\tau_y} - \dfrac{\pi}{2}\right), \end{cases} \tag{14}$$

where L_h is the heat load value, and L_c is the cold load value. $A_h = 6000$ kW is the amplitude of the heat load function, and $A_c = 6000$ kW is the amplitude of the cold load function. $\tau_y = 8760$.

Load balancing constraints

$$\textit{Original constraint}: \ st. \begin{cases} L_h = \eta_{fh}^{B} P_f + \eta_{eh}^{EH} P_{e,EH} + \eta_{eh}^{HP} P_{e,HP}, \\ L_c = \eta_{ec}^{CH} P_{e,CH} + \left(\eta_{eh}^{HP} - 1\right) P_{e,HP}; \end{cases} \tag{15}$$

$$\textit{Relaxation constraint}: \ st. \begin{cases} L_h \le \eta_{fh}^{B} P_f + \eta_{eh}^{EH} P_{e,EH} + \eta_{eh}^{HP} P_{e,HP}, \\ L_c \le \eta_{ec}^{CH} P_{e,CH} + \left(\eta_{ch}^{HP} - 1\right) P_{e,HP}; \end{cases} \tag{16}$$

$$\textit{Loss function}: \ cv. \begin{cases} g_1 = \max\left(0, 1 - \left(\eta_{fh}^{B} P_f + \eta_{eh}^{EH} P_{e,EH} + \eta_{eh}^{HP} P_{e,HP}\right) \Big/ L_h\right), \\ g_2 = \max\left(0, 1 - \left(\eta_{ec}^{CH} P_{e,CH} + \left(\eta_{ch}^{HP} - 1\right) P_{e,HP}\right) \Big/ L_c\right); \end{cases} \tag{17}$$

where $\eta_{fh}^{B} = 0.85$ is the heat coefficient of the natural gas boiler, which represents the heat capacity per unit of natural gas in the natural gas boiler. $\eta_{eh}^{eh} = 0.95$ is the thermal coefficient of the electric boiler, which represents the heat capacity per unit of electric energy in the electric boiler. $\eta_{eh}^{HP} = 6$ is the heat coefficient of the heat pump, representing the heat capacity per unit of electrical energy in the

heat pump. $\eta_{ec}^{CH} = 4$ is the cooling coefficient of the refrigerator, representing the cooling capacity per unit of electrical energy in the refrigerator.

Energy equipment operating constraints

$$st.\begin{cases} P_{e,CH} \geq 0, \\ P_{e,HP} \geq 0, \\ P_{e,EH} \geq 0, \\ P_{f} \geq 0; \end{cases} \tag{18}$$

$$st.\begin{cases} \eta_{ec}^{CH} P_{e,CH} \leq Q_{CH,\max}\ , \\ \left(\eta_{eh}^{HP} - 1\right) P_{e,HP} \leq Q_{HP,\max}\ , \\ \eta_{eh}^{EH} P_{e,EH} \leq Q_{EH,\max}\ , \\ \eta_{fh}^{B} P_{f} \leq Q_{B,\max}\ ; \end{cases} \tag{19}$$

$$st.\ P_{e,HP}(t) \leq \min\left\{ L_c(t) \Big/ (\eta_{eh}^{H} - 1), L_h(t) \Big/ \eta_{eh}^{HP} \right\}; \tag{20}$$

$$Loss\ function:\ g_3 = \max\left(0, 1 - Q_{HP,\max} \Big/ \left(\eta_{eh}^{HP} - 1\right) P_{e,HP}\right); \tag{21}$$

where $Q_{CH,max} \rightarrow \infty$ is the maximum capacity of the refrigerator; $Q_{EH,max} = 2000$ kW is the maximum capacity of the electric boiler, and $Q_{B,max} \rightarrow \infty$ is the natural gas boiler maximum capacity. Based on the working principle of the heat pump, the heat pump can simultaneously heat and cool and the heat capacity of the heat pump is 1.2 times the cooling capacity, so in order not to waste energy, the maximum power of the heat pump does not exceed $\min\left\{ L_c(t)/(\eta_{eh}^{H} - 1), L_h(t)/\eta_{eh}^{HP} \right\}$.

4.3. Solution method

The energy hub system planning problem described in this paper is a mixed-integer programming problem, and it is also a large-scale discrete multiobjective problem. When using the algorithm proposed in this paper to solve the problem, the problem can be divided into two levels: the upper-level optimizer optimizes the capacity of the heat pump, and the lower-level optimizer optimizes the operating power of each energy device. We need to optimize 8760 moments, which obviously affects the computational efficiency. We optimize *n* typical moments and then approximate the optimal solution for atypical moments through the Lagrangian interpolation function. NSGA-II-CDP is used at typical moments. When the optimization of all typical moments is completed and the approximation of atypical moments is also completed, the lower-level optimization results are passed to the upper-level. The upper-level optimizer uses the trisection search method to find a heat pump capacity value and then calls the lower-level optimizer to optimize the operating power of each device under this heat pump capacity value. The upper-level optimizer will frequently call the lower-level optimizer, and the lower-level optimizer will select a solution to pass to the upper-level optimizer through the preference selection mechanism. Then judge through the Pareto dominance relationship, and decide the next search direction of the TSS.

To better describe and explain the algorithm's framework, we give the overall flow chart of the algorithm (Fig. 5). The flowchart is composed of the lower-level optimizer flowchart and the lower optimization flow chart. The main algorithm in the lower-level optimization is NSGA-II-CDP, but the function approximation algorithm is added later. The upper-level optimization is mainly the TSS algorithm, followed by the data collation process. In general, the flow chart expresses the idea and implementation process of the algorithm proposed in this paper well.

4.4. Result analysis

The energy hub system planning problem involved in this paper is a multiobjective problem, and when using the algorithm framework to solve it, a set of non-dominated solutions is obtained. Fig. 6 is the PF obtained by the algorithm framework to solve the problem. When solving practical engineering problems, we usually only need one solution or a few solutions, not the entire PF. Therefore, based on the decision-maker preference information, we give a solution that meets the decision-maker preference requirements. In Fig. 6, domain A indicates that the decision-maker prefers economic benefits, domain B indicates that the decision-maker neither prefers economic benefits nor environmental benefits, and domain C indicates that decision-makers prefer environmental benefits.

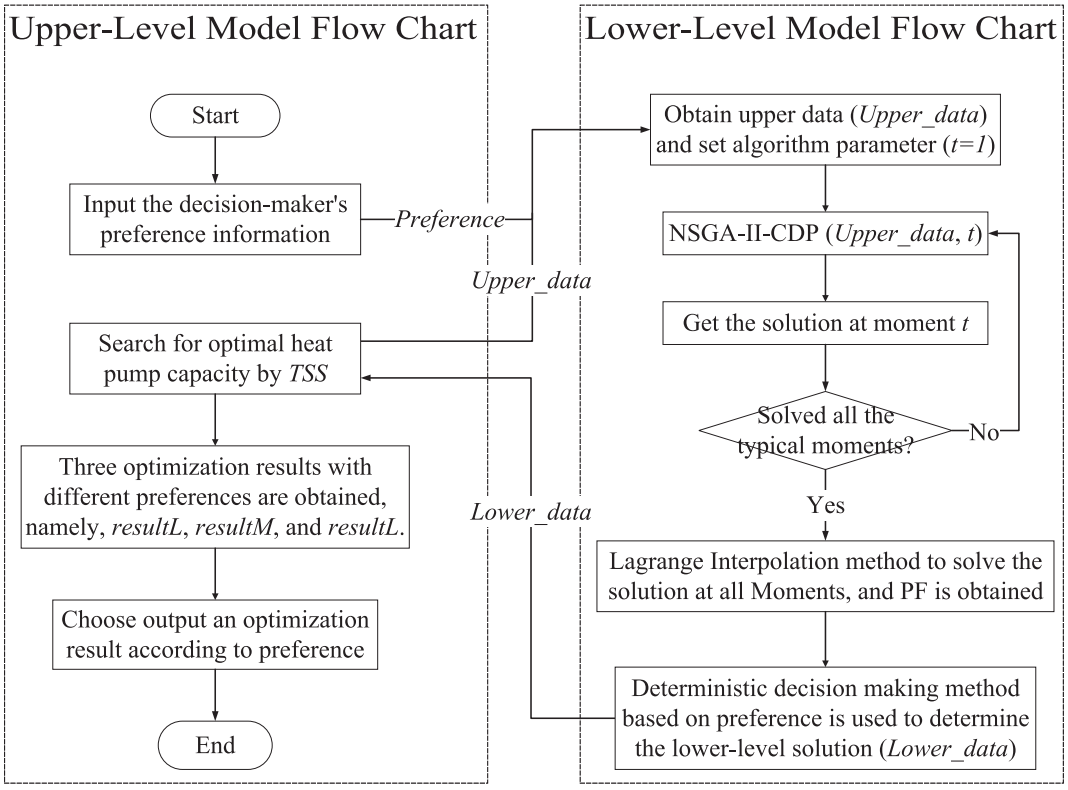


Fig. 5. The overall flow chart of the algorithm, including the flow chart of the upper and lower optimization models. There will be data transmission between the upper-level and lower-level.

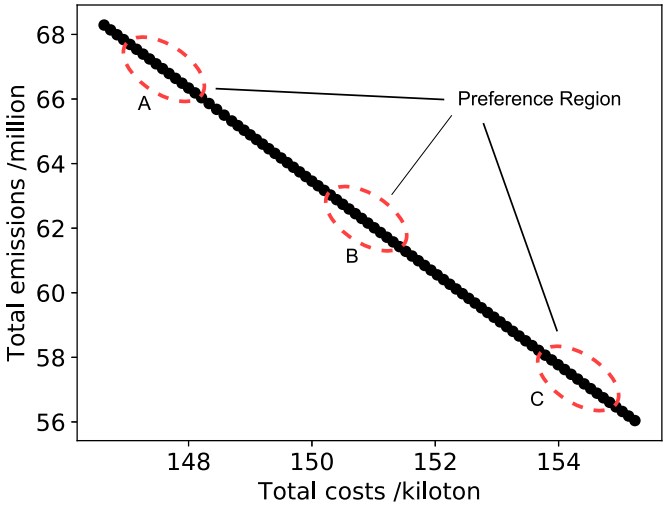


Fig. 6. The PF obtained by using the algorithm proposed in this paper to solve the energy hub system planning problem.

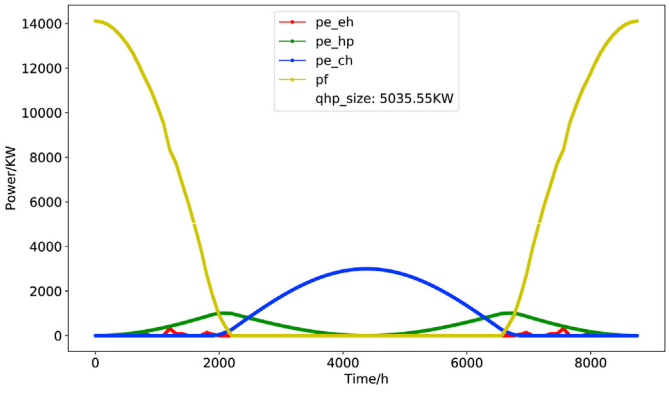


Fig. 7. The year-round optimization solution when the preference area is *A*. The figure shows the electric boiler (*Pe*, *EH*), natural gas boiler (P_f), chillers (*Pe*, *CH*), heat pump (*Pe*, *HP*) real-time power and heat pump capacity ($Q_{HP,\max}$).

If you prefer the total cost of the energy hub system, you can choose the solution of domain A in Fig. 6, please refer to Fig. 7. It can be seen from the figure that in winter, the heat load demand is much greater than the cooling load demand. A large amount of heat load demand is provided by natural gas boilers, and a small amount of cooling load demand is provided by heat pumps. Since the heat pump generates heat when it is cooling, the cooling capacity of the heat pump is exactly equal to the cooling load demand, which is the most economical. In spring and autumn, the energy system makes full use of the heat pump because the demand for cold and heat load in this season is not much different. In summer, the cooling load demand is much greater than the heat load demand. A large

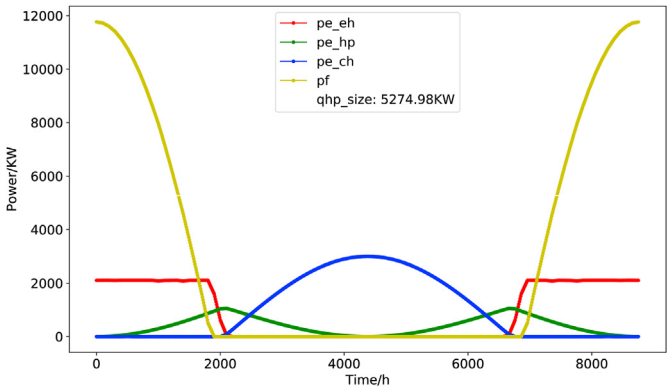


Fig. 8. The year-round optimization solution when the preference area is *C*. The figure shows the electric boiler (*Pe*, *EH*), natural gas boiler (P_f), chillers (*Pe*, *CH*), heat pump (*Pe*, *HP*) real-time power and heat pump capacity ($Q_{HP,\max}$).

amount of cooling load demand is provided by the refrigeration unit, and a small amount of heat load demand is provided by the heat pump. The heating capacity of the heat pump is exactly equal to the heat load demand, which is the most economical. If you prefer the total carbon dioxide emissions of the energy hub system, you can choose the solution of domain C in Fig. 6, see Fig. 8. It can be seen from the figure that compared with the scheme in the Fig. 7, this scheme increases the use of electric boilers in winter. Because the carbon dioxide emissions of electric boilers are less than that of gas-fired boilers, the energy system will maximize the use of electric boilers to provide heat. When the electric boiler cannot meet the thermal load when running at the rated power, the energy system will use gas boilers and heat pumps to supplement heat. If you feel that both the economic and environmental goals are very important and you cannot make decisions, you can choose the solution of domain B in Fig. 6, please refer to Fig. 9. In the figure, you can find that the gas boiler power and electric boiler power curves fluctuate. This is because we need to strike a balance between the economic benefits (low total cost of the energy system) and environmental benefits (low carbon dioxide emissions from the energy system), so energy supply will fluctuate.

The upper-level optimizer obtains different energy hub system planning schemes according to different preferences. We have collated the experimental results (Fig. 10), which shows the objective function values of the three preferences of *LessCost*, *Appropriate*, and *LessExhaust*. It can be seen from Fig. 10 that the solutions of these three preferences are non-dominant. As the total

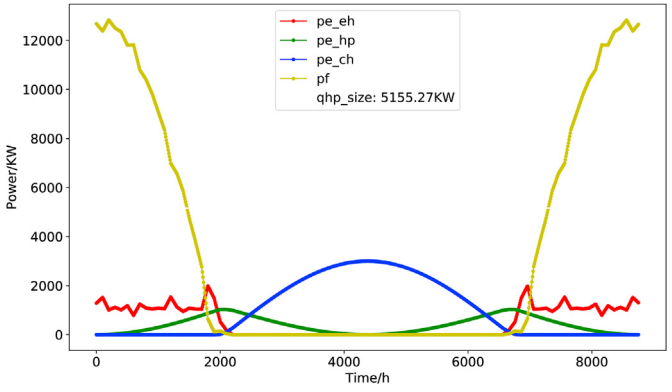


Fig. 9. The year-round optimization solution when the preference area is *B*. The figure shows the electric boiler (*Pe*, *EH*), natural gas boiler (P_f), chillers (*Pe*, *CH*), heat pump (*Pe*, *HP*) real-time power and heat pump capacity ($Q_{HP,\max}$).

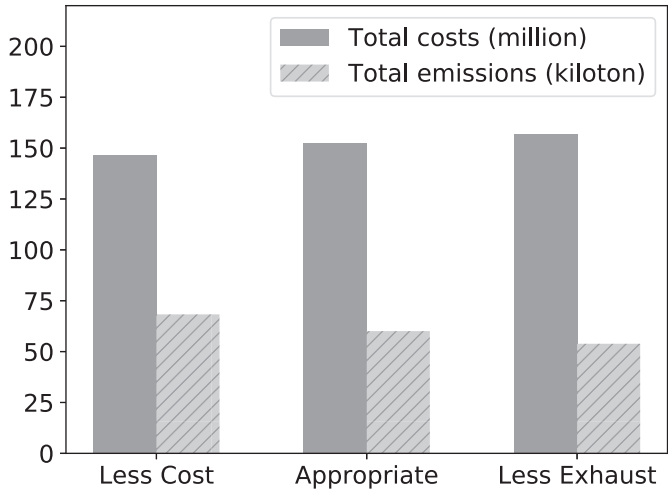


Fig. 10. The objective function values of the three preferences of Less Cost, Appropriate, and Less Exhaust.

cost of the energy hub system decreases, the total emissions of the energy hub system increases. This is consistent with common sense and problem models.

The problem of energy hub system planning is a practical engineering problem, and the correctness and feasibility of the solution are the most important. The algorithm framework proposed in this paper is optimized to obtain three solutions (i.e., *resultL*, *resultM*, *resultR*). We will calculate the optimized result data for analysis (ref Table 1). Since 8760 moments have been optimized, not all of them can be shown in the text, so the table gives some data. From the table, it can be found that the three solutions obtained by optimization are non-dominated. Because electricity is more environmentally friendly than natural gas and electricity is more expensive than natural gas, as the heat pump's capacity increases, the total costs increases, and the total CO_2 emissions decrease. For the optimized solution, it can also be seen that when the heating load demand is high and the cooling load demand is low, the electric boiler and natural gas will be selected. When the heat load demand is low and the cooling load demand is high, the refrigeration unit will be used. When the difference between the cooling and heating loads is small, the energy system will choose the heat pump because the heat pump is the most economical and environmentally friendly. It can be seen from the data in the table that the closer the cooling load to the heating load, the greater the heat pump usage. All data in the table satisfy the model constraints.

5. Conclusions

Energy hub system planning is a discrete large-scale multi-objective optimization problem, so computer deterministic algorithms cannot solve it in a finite time. As the problem's objective function is not differentiable, it isn't easy to use meta-mathematics to solve it. Because the equipment's capacity will affect the equipment's operating power, the optimization of equipment capacity and equipment power is a Stackelberg game process.For these problems, this paper proposes a preference-based multi-objective bilevel optimization algorithm. The lower-level of the algorithm uses evolutionary algorithms to optimize each energy device's real-time operating power in the system. The lower-level can make deterministic decisions through preference information and pass the lower-level solution to the upper-level optimizer. The upper-level uses TSS to search for the best capacity of the device. Finally, this paper selects an energy system case and optimizes it.

Table 1
Heat pump capacity in the energy hub system and real-time power of each equipment in the energy hub system.

Heat pump capacity (kW)	Moment(t)	Heat load(kW)	Cold load(kW)	Electric boiler(kW)	Heat pump(kW)	Chillers(kW)	Natural gas(kW)	Total costs(million yuan)	Total emissions(kiloton)
5035.55	1	12000.00	0.00	0.00	0.00	0.00	14117.65	146.63	68.29
	877	10856.63	1143.37	17.22	229.65	0.00	11127.34		
	1752	7862.29	4137.71	68.27	828.60	0.00	3318.49		
	2628	4154.09	7845.91	0.00	691.82	1097.50	0.00		
	3504	1150.96	10849.04	0.00	191.45	2473.52	0.00		
	4380	0.01	11999.99	0.00	0.02	2999.94	0.00		
	5256	1140.84	10859.16	0.00	190.24	2476.84	0.00		
	6132	4137.71	7862.29	0.00	689.79	1103.10	0.00		
	7008	7845.91	4154.09	72.34	831.05	0.00	3282.16		
	7884	10849.04	1150.96	15.84	230.64	0.00	11116.11		
	8760	12000.00	0.00	0.00	0.00	0.00	14117.65		
5155.27	1	12000.00	0.00	1283.71	0.00	0.00	12683.80	152.39	60.07
	877	10856.63	1143.37	1056.19	229.56	0.15	9968.04		
	1752	7862.29	4137.71	1513.43	828.56	0.07	1708.26		
	2628	4154.09	7845.91	0.00	691.82	1097.50	0.00		
	3504	1150.96	10849.04	0.00	191.45	2473.53	0.00		
	4380	0.01	11999.99	0.00	0.02	2999.95	0.00		
	5256	1140.84	10859.16	0.00	190.24	2476.85	0.00		
	6232	4551.25	7448.75	0.00	758.68	913.65	0.00		
	7008	7845.91	4154.09	1541.95	831.02	0.07	1644.40		
	7884	10849.04	1150.96	1055.33	230.54	0.15	9956.26		
	8760	12000.00	0.00	1283.71	0.00	0.00	12683.80		
5274.98	1	12000.00	0.00	2103.42	0.00	0.00	11767.19	156.83	53.85
	877	10856.63	1143.37	2105.18	229.65	0.27	8793.77		
	1752	7862.29	4137.71	2105.24	828.60	0.00	1050.85		
	2628	4154.09	7845.91	0.00	691.82	1097.52	0.00		
	3504	1150.96	10849.04	0.00	191.45	2473.61	0.00		
	4380	0.01	11999.99	0.00	0.02	2999.94	0.00		
	5256	1140.84	10859.16	0.00	190.24	2476.92	0.00		
	6232	4551.25	7448.75	0.00	758.68	913.66	0.00		
	7008	7845.91	4154.09	2105.24	831.05	0.00	1018.53		
	7884	10849.04	1150.96	2105.19	230.64	0.27	8781.00		
	8760	12000.00	0.00	2103.42	0.00	0.00	11767.19		

The research results show that:

- The lower-level of the algorithm's framework adopts the multiobjective evolutionary algorithm, which is good at solving the problem of the complex objective function and multiobjective NPC problem. Through the case, it was found that the lower-level optimization could accurately obtain the real-time operating power of each device of the energy hub system.
- It is a difficult and important point to solve the problem of non-deterministic decision-making at the lower-level in multi-objective bilevel optimization. The proposed preference selection mechanism can enable the lower-level to make deterministic decisions based on the decision maker's preference information. Therefore, the bilevel optimization algorithm can be perfectly applied in energy system optimization.
- The trisection search proposed in the paper is an algorithm with time complexity $O(log_3 n)$. Through the case, it was found that when the upper-level optimizes the heat pump capacity, the optimal heat pump capacity can be quickly and accurately obtained.

The multiobjective bilevel optimization algorithm based on the preference selection mechanism proposed in the paper can solve the energy hub planning problem and solve a series of similar optimization problems. When the objective function is non-linear or extremely complex, commercial optimizers usually cannot solve it, but the proposed optimization method can solve it. The proposed algorithm has good scalability. When the lower-level optimization problem is a many-objective problem, only the lower-level optimization algorithm needs to be replaced with a many-objective evolutionary algorithm (NSGA-III, MOEA\D, etc.). When the upper-level optimization problem is very complex, the upper-level optimization method can also be changed to an evolutionary algorithm.

Credit author statement

Xu Ynag: Conceptualization, Methodology, Software, Validation, Formal analysis, Resources, Writing-Original Draft, Writing - Review & Editing, Visualization. Juan Zou: Term, Validation, Resources, Supervision, Project administration, Project administration. Zhongbing Liu: Conceptualization, Validation, Investigation, Writing - Original Draft, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. Jiangyang Liu: Validation, Formal analysis, Writing - Original Draft, Writing - Review & Editing. Ling Zhang: Investigation. Jinhua Zheng: Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Authors gratefully appreciate the science and technology project funding from the Headquarters of State Grid Corporation of China (1400-201957438A-0-0-00), the National Natural Science Foundation of China (Grant No. 61876164, 61772178), the Science and

Technology Plan Project of Hunan Province (Grant No. 2016TP1020), the Provinces and Cities Joint Foundation Project (Grant No. 2017JJ4001), and the Hunan province science and technology project funds(Grant No. 2018TP1036).

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